TEST 3—FORM CODE 0Z

SECTION 1 Time—30 minutes 45 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. RECOLLECT: (A) disapprove (B) dispense (C) forget (D) pardon (E) defer
2. EVACUATE: (A) invade (B) find (C) strengthen (D) stabilize (E) delay
3. FALTER: (A) ask (B) lie (C) become famous (D) err repeatedly (E) move steadily
4. DEGENERATION: (A) suppression (B) isolation (C) attention (D) similarity (E) progress
5. ZANY: (A) impious (B) friendly (C) solemn (D) unyielding (E) uninformed
6. SCALE: (A) inflate (B) improve (C) pull apart (D) climb down (E) leave behind
7. RETRACTION: (A) condolence (B) reaffirmation (C) final decision (D) obvious mistake (E) harsh judgment
8. DESTITUTION: (A) activity (B) curiosity (C) maturity (D) prosperity (E) animosity
9. AFFABLE: (A) hardy (B) weary (C) faulty (D) greedy (E) surly
10. OBLIQUE: (A) full (B) permanent (C) direct (D) harmonious (E) unique
11. TENACIOUS: (A) fickle (B) bestial (C) kindly (D) sober (E) ambitious
12. EXTOL: (A) defame (B) alienate (C) conceal (D) refrain (E) reject
13. MERCURIAL: (A) servile (B) stolid (C) prodigal (D) disgruntled (E) obese
14. TORPOR: (A) vivacity (B) coolness (C) aloofness (D) partiality (E) antagonism
15. UPBRAID: (A) loosen (B) praise (C) outwit (D) disguise (E) shorten

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Since we grow soy beans locally, we need not --- them.
(A) feed (B) raise (C) breed (D) import (E) coddle
17. Ironically, the protective law does not cover the type of worker who most obviously --- the protection.
(A) dislikes (B) extends (C) needs (D) opposes (E) exemplifies
18. Schultz cannot stand a situation that is ---, where there is no action.
(A) static (B) chaotic (C) imperfect (D) stressful (E) brutal
19. The people were tired of reform crusades; they wanted no part of an idea that might turn into a ----.
(A) respite (B) reality (C) necessity (D) mistake (E) cause
20. Curiously enough, the ---- of the outcome of the struggle in no way ---- its intensity.
(A) predictability. .condones
(B) uncertainty. .invalidates
(C) ambiguity. .confuses
(D) unimportance. .increases
(E) inevitability. .mitigates

GO ON TO THE NEXT PAGE

Copyright © 1980 by Educational Testing Service All rights reserved.

1

Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

Americans profess a dedication to the cultural ideals of equality of opportunity and the brotherhood of man. The question facing us is whether, in James Baldwin's fine phrase, we shall achieve America.

The 1950's laid the foundation for drastic changes in American race relations. The 1960's were a decade characterized by alterations in the laws governing race relations, the role of race in politics, and even in economic relations between the races. By the mid-1960's the entire legal structure supporting racial segregation and discrimination had been altered. The beginnings of school desegregation had been made in every Southern state, and the federal government had exerted some pressure for more effective desegregation in both faculties and student bodies in Northern city schools. Increased black voter registration in the Southern states and effective political organization and participation of blacks helped to bring about the election of black mayors in Fayette, Mississippi, Gary, Indiana, and Cleveland, Ohio. By 1970 apprenticeships for blacks in the building-trades unions were being negotiated in several cities.

Many white Americans cherish the wistful hope that these gains achieved by black Americans will lessen the pressure for social change. Because the lives of blacks had improved over the last decade, the argument runs, it seems reasonable to expect less militancy, less friction, and reduced interracial conflict. From a social scientist's point of view, it makes better sense to predict increased pressure for equality, more widespread dissatisfaction with discrimination, and more insistent demands for access to the mainstream of American society. Revolutions, whether peaceful or violent, are made not by the most oppressed and downtrodden people in a society, but by those whose position has been improving and who find that access to further improvement is resisted or blocked. The United States became a nation when a revolution was organized by the thirteen best-governed colonies in the world. That revolution was followed by one in France made not by serfs or by an urban proletariat but by the bourgeoisie. The French bourgeoisie in the late eighteenth century had earned a position in some ways analogous to that of many Southern blacks in the 1950's: they had achieved a higher status than the social order was prepared to recognize. The civil rights protest movement of the 1950's and early 1960's was led not by sharecroppers or by unskilled laborers but by clergymen, lawyers, and college students—people who had achieved the occupational and educational levels of middle-class whites but whose new status was not recognized because of their color.

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

21. Which of the following pairs best describes the author's primary purpose in the passage?

(A) To describe and condemn
(B) To analyze and predict
(C) To complain and criticize
(D) To investigate and solve
(E) To boast and encourage

22. The author attributes the election of black mayors in Gary, Indiana, Fayette, Mississippi, and Cleveland, Ohio to which of the following?

I. Changing attitudes of white voters
II. Increased registration of black voters
III. Effective political organization of blacks

(A) II only (B) III only (C) I and II only
(D) II and III only (E) I, II, and III

23. According to the information in the passage, which of the following groups within a society would be most likely to start a revolution?

(A) Those who had once held full privileges in the society but have renounced them
(B) Unskilled laborers who are exploited and unable to enjoy the fruits of their labors
(C) Those who have risen to the middle class but are denied its privileges
(D) Those whose freedoms are severely and unjustly limited by a tyrannical government
(E) Merchants who have earned profits from the exploitation of other groups

24. The author implies that many Americans' devotion to the ideal of the brotherhood of man is

(A) somewhat superficial
(B) unnecessarily optimistic
(C) dangerously fanatical
(D) hopelessly confused
(E) becoming more sincere

25. Which of the following titles best summarizes the content of the passage?

(A) Outlook for Increased Pressure for Equality from Black Americans
(B) Americans Learn the Meaning of "Equal Opportunity"
(C) New Hope for the Minorities of the World
(D) A Summary of the Gains Made by the Civil Rights Movement
(E) Black Americans Take a New Approach to Join the Mainstream of American Society

GO ON TO THE NEXT PAGE

Eighteenth-century physics had required about fifty years from the publication of the Principia to assume with ease the Newtonian posture. Biology has had to traverse even wider confusions before orienting itself cleanly around the Darwinian theory of evolution. Indeed, it has clarified its outlook through the instrumentality of modern genetics only since the 1930's. In 1929 Erik Nordenskjöld's History of Biology, still the foremost text, closed with a reference to "the dissolution of Darwinism." That natural selection "does not operate in the form imagined by Darwin must certainly be taken as proved," wrote Nordenskjöld. Exactly the contrary is now thought to be true. Since such shifts of scientific opinion were still to occur in the twentieth century, it is not surprising that the import of Darwin's theory for the whole science was not mastered at the time. Claude Bernard, perhaps the greatest of experimentalists in his skill and sobriety, saw the future of biology as existing in its reduction of physiology to laws of chemistry and physics. There was nothing for him in Darwin, whose work he did not distinguish from Naturphilosophie. "We must doubtless admire," he writes in his fine manifesto on Experimental Medicine, "those great horizons dimly seen by the genius of a Goethe, an Oken, a Carus, a Geoffroy Saint-Hilaire, a Darwin, in which a general conception shows us all living beings as the expression of types ceaselessly transformed in the evolution of organisms and species,–types in which every living being individually disappears like a reflection of a whole to which it belongs." But he did not admire them. He never thought this theory science. In the critical tradition of French learning, Darwin's mind and language seemed simply slack.

At the other extreme from Gallic indifference burgeoned the enthusiasm which made a religion of science, mistook nature for God, and adopted Darwin as the prophet. The German for this "ism," Darwinismus, best conveys its spirit and was always most at home in Germany. There Ernst Haeckel and his like, deploying all the rich capacity of their language for blurring distinctions, worked a syncretism between the Goethean sense of unity in nature and the Darwinian proof of organic evolution. All biology would be made over into evolution. Thus, to take only one example, embryology was henceforth to be ruled by the doctrine of recapitulation, by which every individual traverses in embryo the evolution of the race, from the single-celled through the invertebrate stage, the gill-breathing, the reptilean, and so on. Everyone, to adapt a saying of Huxley, climbs his own family tree out of the womb.

26. The author mentions Newton's Principia in order to
(A) distinguish between mathematics and biology
(B) demonstrate that new theories are not always valid
(C) compare the scientific attitudes of the eighteenth century with those of the nineteenth century
(D) show that valid original theories are not always immediately accepted by scientists
(E) point out the continuity of science from the eighteenth to the nineteenth century

27. According to the passage, biology did not fully adjust to Darwin's theory until
(A) Nordenskjöld published History of Biology
(B) modern genetics validated his concepts
(C) the doctrine of recapitulation was formulated
(D) the beginning of the twentieth century
(E) embryology was accepted as a valid science

28. The author suggests that in France and Germany scientists
(A) were largely indifferent to Darwin's theories
(B) deployed the capacities of their languages in order to blur distinctions
(C) misunderstood Darwin's theories in different ways
(D) divorced Goethe's theories from those of Darwin
(E) attempted to transform biology into evolution

29. Judging from the passage, one might assume that Ernst Haeckel would have agreed with all of the following statements EXCEPT:
(A) Darwin's theory should not be used to generalize about the nature of man.
(B) Evolution is a process that every organism must undergo.
(C) Goethe's view of nature may be harmonized with that of Darwin.
(D) Embryology obeys the laws of recapitulation.
(E) Biology and evolution are inseparably linked.

30. According to the passage, Haeckel seems to have used Darwin's theory to
(A) explain only human biology
(B) refute the claims that there is unity in nature
(C) reinforce the use of scientific method
(D) condemn those who confuse nature with God
(E) make very broad philosophical claims

GO ON TO THE NEXT PAGE

1

Select the word or set of words that best completes each of the following sentences.

31. Edwards had an ---- thirst for honor and glory, as though the balm of innumerable tributes could never quite --- the wounds he suffered in those long years of failure and neglect.

(A) unquenchable. .heal (B) envious. .magnify
(C) inconsequential. .avenge (D) awesome. .shatter
(E) insufficient. .soothe

32. In her description of the brain as an interpreter of messages from the nervous system, Henderson cautioned that interpreters, however proficient, are --- and can ---- communications.

(A) skillful. .disclose (B) instructive. .convey
(C) incompetent. .impart (D) articulate. .garble
(E) fallible. .distort

33. As most biographers have learned, the relationships between an artist's life and his or her work can be so --- and complicated that they ---- the most determined scrutiny.

(A) tedious. .enliven (B) momentous. .nullify
(C) elusive. .resist (D) conspicuous. .defy
(E) fascinating. .discourage

34. The --- of governmental power makes it impossible for a potential ---- to know whom to flatter.

(A) benignity. .statesman
(B) ambiguity. .sycophant
(C) changeability. .rebel
(D) rigidity. .member
(E) morality. .believer

35. Unless the special interests of a biological species demand --- of its members, it is obviously --- to spread the individuals as evenly as possible over the available habitat.

(A) colonization. .demonstrative
(B) decentralization. .convenient
(C) minimization. .inefficient
(D) aggregation. .expedient
(E) concentration. .ineffective

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. SANDPAPER : ABRASIVE ::
(A) disinfectant : contagious (B) ointment : painful
(C) glue : sticky (D) money : expensive
(E) muffler : noisy

37. POLKA-DOTTED : SPOTS :: (A) transparent : holes
(B) speckled : threads (C) knotted : fabrics
(D) striped : lines (E) plaid : triangles

38. PLANE : HANGAR :: (A) coat : closet
(B) house : garage (C) food : store
(D) submarine : water (E) traffic : highway

39. ECHO : LISTENER :: (A) microphone : speaker
(B) leverage : lifter (C) reflection : viewer
(D) rehearsal : actor (E) monotony : interrupter

40. REINDEER : TUNDRA :: (A) bear : den
(B) pony : stable (C) sheep : sheepfold
(D) mustang : corral (E) camel : desert

41. EMACIATED : NOURISHMENT ::
(A) rumpled : clothing (B) neurotic : nerve
(C) cured : medicine (D) poisoned : hunger
(E) flabby : exercise

42. BLUNDERBUSS : FIREARM :: (A) galleon : vessel
(B) lance : spear (C) parachute : aircraft
(D) khaki : uniform (E) sword : dagger

43. COURAGEOUS : FOOLHARDY :: (A) alert : severe
(B) modest : slovenly (C) generous : corrupt
(D) innocent : liberal (E) unpleasant : noxious

44. EXORCISM : DEMON :: (A) divorce : infidelity
(B) absolution : guilt (C) convocation : angel
(D) displacement : traveler (E) revelation : spirit

45. PREVARICATION : LIAR :: (A) banality : bore
(B) blasphemy : idol (C) idiom : imbecile
(D) incantation : soloist (E) compliment : egotist

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 2 Time—30 minutes, 25 Questions

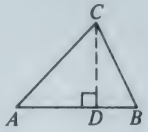
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

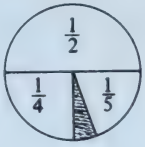
(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. Which of the following must be even?

   I. The sum of two odd numbers
   II. The sum of an odd and an even number
   III. The sum of two even numbers

   (A) I only (B) II only (C) I and II only
   (D) I and III only (E) I, II, and III



2. In the figure above, the shaded region is what fractional part of the area of the circle?

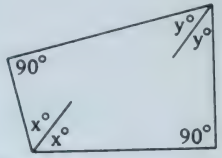
   (A) $\frac{1}{60}$ (B) $\frac{1}{40}$ (C) $\frac{1}{20}$ (D) $\frac{1}{10}$ (E) $\frac{1}{6}$

3. If $P_1 = 4 \times 0.095$ and $P_2 = 0.004 \times 0.095$, which of the following is true?

   (A) $P_1 = P_2$
   (B) $P_2 > P_1$
   (C) $P_1 + P_2 > 5$
   (D) $P_1 > 4$
   (E) $P_1 > P_2$

4. If $x + 3 = x - b$, then $b =$

   (A) -3 (B) 3 (C) $-x$ (D) x (E) $-2x + 3$



5. In the figure above, $x + y =$

   (A) 30 (B) 45 (C) 60 (D) 90 (E) 120

6. How many numbers between 20 and 50 are each equal to 9 times an integer?

   (A) One
   (B) Two
   (C) Three
   (D) Four
   (E) Nine

7. If $(r + 1)\left(\frac{1}{r}\right) = 0$, what is r?

   (A) 2 (B) 1 (C) -1
   (D) -2 (E) Any integer

GO ON TO THE NEXT PAGE

2

8. If x, y, and z are consecutive integers and $x + y + z = 270$, what is the value of the LEAST of the three integers?

   (A) 80 (B) 89 (C) 90 (D) 268

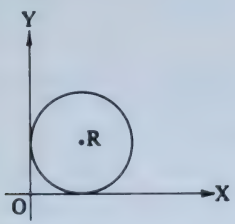
   (E) It cannot be determined from the information given.

9. If the average of 2 numbers is 7 and the product of the 2 numbers is 48, then the positive difference between the 2 numbers is

   (A) 1
   (B) 2
   (C) 3
   (D) 4
   (E) 6

10. If $3 \cdot 5 \cdot 7 = x$ and $3 \cdot 5 = y$, then $x - y$ is equal to the product of 15 and

    (A) 7 (B) 6 (C) 1 (D) -6 (E) -7



11. In the figure above, a circle with center R has an area of 36π square units and just touches the X- and Y-axes. What are the coordinates of R?

    (A) (3, 12) (B) (4, 9) (C) (6, 6)

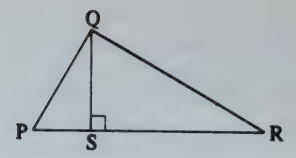
    (D) (18, 18) (E) (36, 36)

---

12. If $\frac{1}{x + y} = 1$, then $x =$

    (A) $1 - y$ (B) $y - 1$ (C) $\frac{1}{1 + y}$

    (D) $1 + y$ (E) $-y$



Note: **Figure not drawn to scale.**

13. In $\triangle PQR$ above, $QS \perp PR$. Which of the following conditions will insure that area $\triangle PQS <$ area $\triangle SQR$?

    (A) $QS < PQ$ (B) $PQ < PR$ (C) $QR < PR$

    (D) $PQ < QR$ (E) $QS < PR$

---

14. If 3 persons who work at the same rate can do a job together in 5 days, what fractional part of that job can one of these persons do in 1 day?

    (A) $\frac{1}{15}$ (B) $\frac{1}{12}$ (C) $\frac{1}{9}$ (D) $\frac{1}{5}$ (E) $\frac{1}{3}$



15. If the pattern of the first 5 arrows above continues to the right, the 107th arrow would be in which of the following directions?

    (A) ↓ (B) ↘ (C) ↗ (D) ↑ (E) →

Row A	7	2	5	4	6
Row B	3	8	6	9	7
Row C	5	4	3	8	2
Row D	9	5	7	3	6
Row E	5	6	3	7	4

16. Which row in the list above contains both the square of an integer and the cube of a different integer?

    (A) Row A (B) Row B (C) Row C

    (D) Row D (E) Row E

GO ON TO THE NEXT PAGE

17. Jack begins reading at the top of page N and finishes at the bottom of page R. If the pages are numbered and read consecutively and if there are no blank pages, how many pages has he read?

(A) R - N + 1
(B) N - R + 1
(C) N - R - 1
(D) R - N
(E) N - R

18. Out of a total of 154 games played, a ball team won 54 more games than it lost. If there were no ties, how many games did the team win?

(A) 94
(B) 98
(C) 100
(D) 102
(E) 104

19. On a scale drawing, if 4 centimeters represent 16 meters, how many centimeters represent 2 meters 8 centimeters?
(100 centimeters = 1 meter)

(A) 0.52
(B) 0.58
(C) 0.70
(D) 2.08
(E) 2.50

20. If x is k per cent of y, what per cent of y is kx ?

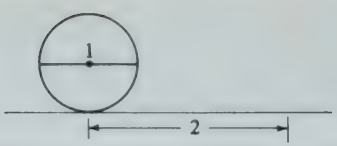
(A) $\frac{k}{100}\%$

(B) $\frac{100}{k}\%$

(C) k%

(D) 100k%

(E) $k^2\%$



21. The figure above shows a circle with diameter 1. If $\frac{22}{7}$ is used as an approximation to π and if the circle rolls a distance 2, what part of a complete rotation does it make?

(A) $\frac{7}{44}$

(B) $\frac{7}{22}$

(C) $\frac{7}{11}$

(D) $\frac{11}{7}$

(E) $\frac{22}{7}$

22. At a cost of 5 oranges for t cents, how many oranges can be bought for x dollars?

(A) $\frac{500x}{t}$

(B) $\frac{500t}{x}$

(C) $\frac{20x}{t}$

(D) $\frac{20t}{x}$

(E) 20tx

GO ON TO THE NEXT PAGE

2

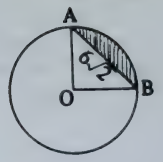
23. A triangle is obtuse if and only if the degree measure of one of its angles is greater than 90 and less than 180. A certain triangle has sides of lengths 5, 5, and x. What are all numbers x for which this triangle is obtuse?

(A) $5 \leqq x \leqq 5\sqrt{2}$
(B) $5 < x < 5\sqrt{2}$
(C) $5\sqrt{2} \leqq x \leqq 10$
(D) $5\sqrt{2} < x \leqq 10$
(E) $5\sqrt{2} < x < 10$

$$\begin{array}{r} A\,B \\ +C\,D \\ \hline A\,A\,A \end{array}$$

24. In the correct addition problem shown above, A, B, C, and D represent nonzero digits. What is the value of C ?

(A) 9 (B) 8 (C) 7 (D) 1
(E) It cannot be determined from the information given.



25. In the figure above, if O is the center of the circle and the length of side AB of right triangle AOB is $6\sqrt{2}$, what is the area of the shaded region?

(A) $27\pi + 18$
(B) $27\pi - 18$
(C) $9\pi + 18$
(D) $9\pi - 18$
(E) It cannot be determined from the information given.

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 3 Time—30 minutes 40 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. CONFIDENTIAL: (A) baffling (B) amusing (C) straightforward (D) widely known (E) fully understood

2. ACCENT: (A) lack of emphasis (B) loss of face (C) knack of writing (D) difficulty in hearing (E) promptness in replying

3. EMERGE: (A) reduce (B) disappear (C) float (D) collide (E) puncture

4. MANIPULATABLE: (A) creative (B) inflexible (C) methodical (D) inconvenient (E) exaggerated

5. DUBIOUS: (A) early (B) lovely (C) distinct (D) easily forgiven (E) rapidly developing

6. EFFACE: (A) trust (B) restore (C) embody (D) contract (E) appreciate

7. GLACIAL: (A) hot (B) placid (C) smooth (D) finite (E) protected

8. DECOROUSNESS: (A) lack of propriety (B) lack of respect (C) faulty perception (D) extreme simplicity (E) uninspired performance

9. AMBULATORY: (A) irritable (B) satisfactory (C) hospitalized (D) self-propelled (E) bedridden

10. TRACTABLE: (A) willful (B) intact (C) delectable (D) irksome (E) obscure

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. With modern equipment it is now possible to tell how far away from the observatory the earthquake ----; three observatories acting together can ---- the center of the quake.

(A) traveled. .reconstruct (B) scattered. .calculate (C) originated. .pinpoint (D) ascended. .locate (E) arose. .probe

12. Clichés abound not only in the writing style of this volume but also in its surpassingly ---- illustrations.

(A) subtle (B) inventive (C) moving (D) trite (E) uncertain

13. Food and sex are very basic requirements of animal life, but certain hermits eschewed sexual activity altogether and ---- food consumption to the lowest point ---- survival.

(A) limited. .injurious to
(B) cultivated. .vulnerable to
(C) reduced. .compatible with
(D) devoured. .recorded for
(E) expanded. .helpful for

GO ON TO THE NEXT PAGE

14. Even though this book has some ---- pages, even though it rarely breaks into eloquence or ----, the story it tells is nevertheless a deeply moving one.
    (A) insipid. .drivel
    (B) prosaic. .profundity
    (C) memorable. .credibility
    (D) tedious. .superficiality
    (E) insightful. .enlightenment

15. The ---- he exhibits in financial dealings is mirrored in his personal relationships; never have I seen a man so ---- kind words and considerate deeds.
    (A) extravagance. .oblivious to
    (B) haughtiness. .grateful for
    (C) parsimony. .chary of
    (D) rancor. .generous with
    (E) benevolence. .covetous of

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep (B) anger : madness (C) smile : amusement (D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. WORD : SENTENCE :: (A) signature : photograph (B) question : response (C) paragraph : essay (D) stamp : envelope (E) applause : performance

17. DUCK : DECOY :: (A) dog : bone (B) bee : honey (C) horse : fodder (D) insect : flyswatter (E) fish : bait

18. SLOUCH : POSTURE :: (A) scribble : penmanship (B) hum : music (C) yawn : rest (D) relax : formality (E) confuse : logic

19. LOOM : WEAVER :: (A) gallery : painter (B) mobile : sculptor (C) nail : architect (D) design : potter (E) anvil : blacksmith

20. STALE : BREAD :: (A) mellow : age (B) skimmed : milk (C) sour : taste (D) rancid : butter (E) fermented : yeast

21. RED TAPE : EFFICIENCY ::
    (A) white gold : value
    (B) bluestocking : poetry
    (C) brownstone : building
    (D) soft soap : flattery
    (E) wet blanket : enthusiasm

22. APATHETIC : EMOTION ::
    (A) bored : leisure
    (B) hypothetical : imagination
    (C) controversial : anger
    (D) dramatic : quality
    (E) neutral : bias

23. STOKE : FIRE :: (A) bake : oven (B) laugh : joke (C) incite : crowd (D) declaim : stage (E) hoard : money

24. GULLIBILITY : CHARLATAN :: (A) bondage : slave (B) fear : bully (C) courage : conspirator (D) desire : optimist (E) submissiveness : follower

25. CORROSION : METAL :: (A) lethargy : action (B) atrophy : body (C) illness : vision (D) expansion : business (E) drought : land

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

The land has housed us, clothed us, and fed us, but again and again, out of indifference or sheer stupidity, we have squandered resources, defiled, destroyed, and moved on. The land was so rich, so abundant, so very good, we believed, that there was just no end to its capacity to produce or to recover from our mistakes. "There is always more where that came from," we said.

Now, however, in the last third of the twentieth century, we are thinking differently. Our waste—our cutting down of entire forests, our poisoning of air and water, our mindless interference with the vital life cycles of plants and animals—is catching up with us. Although our productivity and our prosperity grow steadily, scientists keep warning of the terrible day of reckoning. As any thoughtful person realizes, we Americans and our land have reached a stage of crisis.

Future historians will no doubt point to this as the time when we first traveled far enough toward the stars to see with our own eyes our earth-home whole: a mere speck in space, but still our only home and the only source of life we know anything about.

26. All of the following are specifically mentioned in the passage as misuses of the Earth's resources EXCEPT

(A) interfering with the life cycle of plants and animals
(B) poisoning the water
(C) squandering our natural resources
(D) destroying our forests
(E) ruining the ecological balance of the oceans

27. In describing American attitudes about the land (lines 7-8), the author implies that

(A) people were shortsighted
(B) people's desires were limited
(C) America's resources were inexhaustible
(D) America has always been short on resources
(E) individuals must look out for their own best interests

28. According to the passage, for which of the following reasons did we exploit the land?

I. Carelessness
II. Insensitivity
III. A desire to conquer the land in order to move the frontier forward

(A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II, and III

GO ON TO THE NEXT PAGE

3

Animals are not moved by what they cannot react to. They live in a tiny world, a sliver of reality that keeps them walking behind their noses and shuts out everything else. But look at humans. Here nature created an animal with no defense against full perception of the external world, an animal completely open to experience. Humans relate not only to others in their own species, but to animals in all other species. They can contemplate not only what is edible for them, but everything that grows. They live not only in this moment, but they expand their inner selves to yesterday, their curiosity to years ago, their fears to five billion years from now when the sun will cool, their hopes to an eternity. They live not on a tiny territory, nor even on a planet, but in the universe. The burden is appalling. Humans cannot even take their bodies for granted as can other animals. Their limbs are not just "there," to be used and taken for granted. Not only is the body strange, but also its inner landscape, the memories and dreams. They do not know who they are or why they are born. Their existence is incomprehensible to them.

Line (5) (10) (15) (20)

29. The main point of the passage is to

(A) show the development of human intelligence
(B) emphasize the problems that awareness brings to humans
(C) indicate the superiority of the human race over other animal species
(D) contrast the inner world of humans to the external world in which they live
(E) explain why animals react to stimuli the way they do

30. The author describes the world of animals as "a sliver of reality that keeps them walking behind their noses and shuts out everything else" (lines 2-4) in order to

(A) establish the difference between reality and the perception of reality
(B) demonstrate how much animals depend on their senses for survival
(C) show how animals react to members of their own species
(D) explain why strange stimuli are so frightening to animals
(E) exemplify the limitations of an animal's perception

31. The passage suggests that both animals and people are capable of

(A) dreaming
(B) curiosity
(C) living in a limited sensory environment
(D) reacting to others in their own species
(E) taking their bodies for granted

32. Which of the following most fully represents the "burden" referred to in line 17 ?

(A) "a sliver of reality" (lines 2-3)
(B) "full perception of the external world" (line 6)
(C) "five billion years from now when the sun will cool" (lines 14-15)
(D) "their bodies" (line 18)
(E) "memories and dreams" (lines 21-22)

GO ON TO THE NEXT PAGE

A scientific theory is a step—in both senses of the word—toward understanding. A useful theory is in itself an advancement of our knowledge, and it is also a firm base for further advancement. Scientific theories are intellectual constructs that relate broad collections of experimentally observed facts. The advancement of knowledge inherent in a theory is in this relating or "explaining" of otherwise isolated and trivial sets of facts. The base for further advancement is in prediction, guidance of new experiments, and extension to new and broader theories. The basis, existence, and function of a scientific theory are pervaded by experimental observation.

Here, then, is why the atomic theory of the ancients cannot be considered a scientific theory in the modern sense. It was not based on observation, nor did it lead to experiments or further theories. Indeed, the notion of "ultimate particle" discouraged both experiment and speculation beyond this point. As long as the existence of atoms was postulated only because it seemed fitting, beautiful, moral, or God's will, no science had been created, and no understanding had been approached. But when John Dalton, in 1804, postulated an atomic structure because matter behaves in such a way that it must be particulate, our understanding took a long and bold step forward. The atomic theory became a scientific theory.

33. Which of the following best describes the content of the passage?

(A) A description of the first atomic theory
(B) An explanation of the characteristics of a scientific theory
(C) Advice to students concerning the application of scientific theory
(D) A summary of events leading to Dalton's theory of atomic structure
(E) A condemnation of present confusion about the nature of scientific theories

34. Which of the following pairs of words best represents what the author means by "both senses" of the word "step" (lines 1-2) ?

(A) Space and size
(B) Challenge and rebuttal
(C) Imagination and success
(D) Foundation and progression
(E) Experiment and observation

35. It can be inferred that the ancients' atomic theory was based primarily on

(A) previous scientific theories
(B) primitive scientific experiments
(C) philosophical or theological judgments
(D) observations of isolated facts
(E) attempts to stimulate new experimentation

GO ON TO THE NEXT PAGE

3

"Well, if you wrote her about my illness, we are in a fix," the poor man said in a resigned tone. "Let's see if we can get out of it. Take a telegraph blank and a pencil, and let us compose a diplomatic dispatch." And slowly, with a quizzical smile playing about the corners of his mouth, he began to dictate: "Father better. Leaving as soon as possible for south of France with good nurse. I sail for home September first. Don't undertake voyage yourself. Utterly useless." He perused the telegram with caution. "We shall amplify one sentence," he observed judiciously. "We will say, 'Please don't undertake long voyage yourself.' That will sound more as if you had sent the telegram and not I."

Those two polite considerate words were not enough to deceive the practiced eye of Mrs. Alden. She saw that it was the Doctor's effort to keep her away. He was probably not better, except that he might have had worse moments. When the first of September came he would be worse again, and would keep Oliver from sailing. Her poor boy's feelings would be worked upon. He would be told not to abandon his dying father. And this dying father would live on, drag him to Marseilles, carry him away in that ill-fated yacht, cause him to miss his college year, and subject him to the evil influence of that wicked young Captain. Hadn't she employed a detective agency to look up the man's past and his present, and hadn't they discovered the most appalling facts, though nothing that she hadn't already known by instinct? It was a wicked plot to destroy whatever conscience and sense of decency might still remain in this weak, this sentimental boy. She would take the next steamer for Liverpool and defeat that conspiracy. She would sail from Boston, because the Boston boats would surely be safer; the Cunard line was of course the only right one—Letitia Lamb always traveled by it. Her reply to Oliver's cablegram was brief: "Sail Saturday Lucania." It was truer, she was happy to think, than the long one she had received, less expensive, and more expressive of a strong, upright, determined, unselfish character.

36. In this story, the most important aspect of the relationship between the Doctor and his wife is probably their

(A) underlying affection for one another
(B) rivalry for control of their son
(C) lack of concern for the welfare of the other
(D) refusal to disguise their deception of each other
(E) disagreement concerning the value of a college education

37. It can be inferred that Mrs. Alden's attitude toward Letitia Lamb is one of

(A) admiration (B) curiosity (C) distrust (D) hypocrisy (E) jealousy

38. Mrs. Alden thinks that the brevity of her telegram is most indicative of her

(A) need to economize
(B) fear of deception
(C) honest concern
(D) wrathful indignation
(E) forceful personality

39. The Doctor's primary purpose in changing the original draft of the telegram is to

(A) mislead Mrs. Alden into believing that he is concerned about her welfare
(B) mislead Mrs. Alden into believing that her son has written the message
(C) exaggerate the actual severity of his illness
(D) emphasize that he really wants Mrs. Alden to stay away
(E) disguise his real attitude toward Mrs. Alden

40. On the basis of the passage, the author's attitude toward Mrs. Alden can most accurately be termed one of

(A) scornful indifference
(B) sympathetic concern
(C) overt disapproval
(D) detached amusement
(E) sentimental involvement

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 5 Time—30 minutes 35 Questions

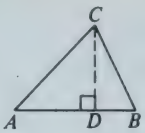
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

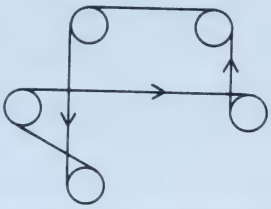
1. $(2^3 - 2^4)(2^2 - 2) =$

(A) 8 (B) 0 (C) -4 (D) -8 (E) -16



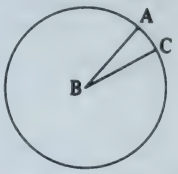
2. In the figure above, if PQ is a line segment, then x =

(A) 90
(B) 60
(C) 45
(D) 30
(E) 15



3. In the figure above, a belt which runs over five wheels moves in the direction of the arrows. How many of the wheels are turning clockwise?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5



4. In the figure above, the length of the minor arc AC is $\frac{1}{18}$ of the circumference of the circle with center B. What is the measure of $\angle ABC$ in degrees?

(A) 30° (B) 25° (C) 20° (D) 10° (E) 5°

30, 31, 32, 33, 34, 35

5. If S is the sum of the remainders when each number in the list above is divided by 6, what is the remainder when S is divided by 6 ?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 5

6. The average of positive numbers x, y, and z is what per cent of their sum?

(A) 25% (B) $33\frac{1}{3}\%$ (C) 50% (D) $66\frac{2}{3}\%$

(E) It cannot be determined from the information given.

7. If 70 tickets to a play were bought for a total of \$50.00 and if tickets cost \$1.00 for adults and \$0.50 for children, how many children's tickets were bought?

(A) 20 (B) 25 (C) 30 (D) 35 (E) 40

GO ON TO THE NEXT PAGE

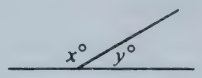
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet fill in oval

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

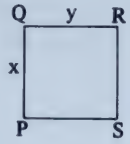
EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ Ⓔ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ Ⓔ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ● Ⓔ

Notes:

1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

	Column A	Column B
8.	$\frac{1}{2} + \frac{1}{3}$	$\frac{1}{4} + \frac{1}{5}$



PQRS is a square.

9.	x^2	y^2

$9 - 3x = 6$
$12 - 2y = 10$

10.	x	y
11.	$\sqrt{2} - 1$	$\sqrt{3} - 1$

In a certain class, every student studies either art or science or both; 12 study only art, 10 study only science, and x study both.

12.	Number of students in the class who study art but not science	Number of students in the class who study science but not art

	Column A	Column B
13.	The perimeter of a triangle	The perimeter of a rectangle
14.	$1 + \dfrac{1}{1 + \frac{1}{2}}$	2
15.	The number of articles X can produce per hour if he produces at the rate of 25 articles in 15 minutes	The number of articles Y can produce per hour if he produces at the rate of 20 articles in 12 minutes

$x > 0$

16.	$\sqrt{x + 1}$	$\sqrt{x} + 1$

k is an integer greater than 1.

17.	$k^2(k - 1)^2$	36

$P = (x - 2)(x + 3)$
$Q = (x - 3)(x + 2)$

18.	Value of P when x = 2	Value of Q when x = 2

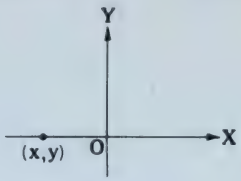
GO ON TO THE NEXT PAGE

SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

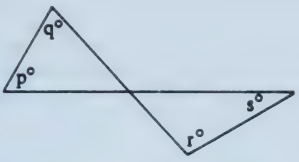
AN E RESPONSE WILL NOT BE SCORED.

	Column A	Column B

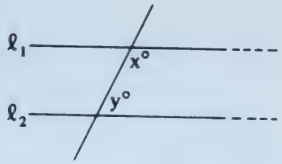


Rectangular Coordinates

19. x y



20. p + q r + s



Lines ℓ_1 and ℓ_2 meet when extended to the right.

21. x + y 180

A coat that was priced at $36.50 is sold at a 30 per cent discount.

22. Price of the coat after discount $25.55

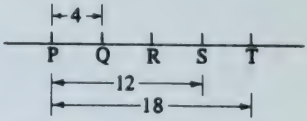
$x < 0$
$y > 1$

23. xy $\frac{x}{y}$

	Column A	Column B

In a game, a fair coin is to be tossed 10 times and the score will be the number of heads obtained.

24. The chances of a score of exactly 7 | The chances of a score of exactly 3



Note: Figure not drawn to scale.

25. ST RS

BEEF PRICES

Brand	Price per Pound
X	$1.40
Y	$1.20

26. Average price per pound of 2 pounds of brand X beef and 3 pounds of brand Y beef | Average price per pound of 2 pounds of brand X beef and 1 pound of brand Y beef

---

$x < 0$

27. $(x + 2)(x + 3)$ $(x + 2)^2$

GO ON TO THE NEXT PAGE

5

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

28. If $6t - p = 8$ and $4t - 3p = 7$, then $t + p =$

(A) $\frac{1}{2}$ (B) 1 (C) $\frac{17}{14}$ (D) $\frac{27}{14}$ (E) 3

Questions 29–30 refer to the following definition.

For all real numbers p and q, where $q \neq 0$,

$$\left(\frac{p}{q}\right) = \frac{p^2}{q^2}$$

29. $\left(\frac{3}{5}\right)\left(\frac{5}{6}\right) =$

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4 (E) 8

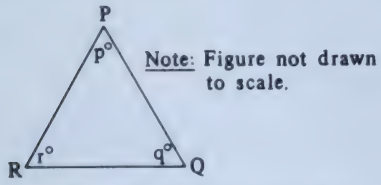
30. If $pqrs \neq 0$, which of the following is (are) necessarily true?

I. $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$

II. $\left(\frac{p}{q}\right)\left(\frac{\frac{1}{p}}{\frac{1}{q}}\right) = 1$

III. $\left(\frac{p}{q}\right)\left(\frac{r}{s}\right) = \left(\frac{pr}{qs}\right)$

(A) I only
(B) I and II only
(C) I and III only
(D) II and III only
(E) I, II, and III



Note: Figure not drawn to scale.

31. In ΔPQR above, if $PQ < RQ < PR$, then

(A) $q < r$ (B) $p < r$ (C) $q < p$
(D) $p < 60$ (E) $r < 60$

32. A person is paid time and a half for all hours worked in excess of 7.5 hours per day. If the person works 10 hours in one day, by what per cent are his regular wages for the day increased?

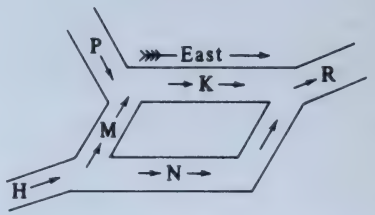
(A) 10% (B) 20% (C) 25%
(D) $33\frac{1}{3}\%$ (E) 50%

33. A closed rectangular tank 1 meter by 2 meters by 4 meters contains 4 cubic meters of water. When the tank is placed level on its various sides, the water depth changes. What is the greatest possible difference in water depths?

(A) 0.5 m
(B) 1 m
(C) 1.5 m
(D) 2 m
(E) 3 m

34. If the average of the degree measures of two angles of an isosceles triangle is 50, which of the following could be the degree measure of one of the angles of the triangle?

(A) 20 (B) 40 (C) 60 (D) 90
(E) It cannot be any of the above.



35. In the highway system illustrated above, one-tenth of the eastbound cars from highway H turn into M; the rest continue on N. One-fifth of the eastbound cars on K come from P; the rest, from M. If traffic on all highways is in the direction of the arrows, what is the ratio of the traffic on H to that on R?

(A) 1 to 2
(B) 1 to 8
(C) 5 to 7
(D) 5 to 8
(E) 40 to 41

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

Correct Answers for Scholastic Aptitude Test
Form Code 0Z

VERBAL		MATHEMATICAL	
Section 1	Section 3	Section 2	Section 5
1. C	1. D	1. D	1. E
2. A	2. A	2. C	2. D
3. E	3. B	3. E	3. B
4. E	4. B	4. A	4. C
5. C	5. C	5. D	5. D
6. D	6. B	6. C	6. B
7. B	7. A	7. C	7. E
8. D	8. A	8. B	*8. A
9. E	9. E	9. B	*9. C
10. C	10. A	10. B	*10. C
11. A	11. C	11. C	*11. B
12. A	12. D	12. A	*12. A
13. B	13. C	13. D	*13. D
14. A	14. B	14. A	*14. B
15. B	15. C	15. B	*15. C
16. D	16. C	16. B or C	*16. B
17. C	17. E	17. A	*17. D
18. A	18. A	18. E	*18. A
19. E	19. E	19. A	*19. B
20. E	20. D	20. E	*20. C
21. B	21. E	21. C	*21. B
22. D	22. E	22. A	*22. C
23. C	23. C	23. E	*23. B
24. A	24. B	24. A	*24. C
25. A	25. B	25. D	*25. D
26. D	26. E		*26. B
27. B	27. A		*27. D
28. C	28. B		28. A
29. A	29. B		29. A
30. E	30. E		30. D
31. A	31. D		31. E
32. E	32. B		32. E
33. C	33. B		33. C
34. B	34. D		34. A
35. D	35. C		35. E
36. C	36. B		
37. D	37. A		
38. A	38. E		
39. C	39. B		
40. E	40. D		
41. E			
42. A			
43. E			
44. B			
45. A			

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.

- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

SAT-Verbal Sections 1 and 3

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 122. Enter this figure on the worksheet.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 122. Enter this figure on the worksheet.

FORM CODE 0Z

SAT SCORING WORKSHEET

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____ (no. incorrect)) = _____ subtotal A

B. Section 3: _____ (no. correct) − ¼ (_____ (no. incorrect)) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on page 122.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ (no. correct) − ¼ (_____ (no. incorrect)) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____ (no. incorrect)) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____ (no. incorrect)) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on page 122.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 0Z

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	470	600
84	790		39	460	590
83	780		38	460	590
82	770		37	450	580
81	760		36	440	570
80	750		35	440	560
79	740		34	430	550
78	730		33	420	540
77	730		32	420	530
76	720		31	410	530
75	710		30	400	520
74	700		29	400	510
73	690		28	390	500
72	680		27	380	490
71	680		26	380	480
70	670		25	370	470
69	660		24	360	470
68	660		23	360	460
67	650		22	350	450
66	640		21	340	440
65	640		20	340	430
64	630		19	330	420
63	620		18	320	410
62	620		17	320	410
61	610		16	310	400
60	600	800	15	310	390
59	600	790	14	300	380
58	590	780	13	290	370
57	580	770	12	290	360
56	580	760	11	280	350
55	570	750	10	270	350
54	560	740	9	270	340
53	560	730	8	260	330
52	550	720	7	250	320
51	540	710	6	250	310
50	540	700	5	240	300
49	530	690	4	230	290
48	520	680	3	230	290
47	520	670	2	220	280
46	510	660	1	210	270
45	500	650	0	210	260
44	500	640	– 1	200	250
43	490	630	– 2	200	240
42	480	620	– 3	200	230
41	480	610	– 4	200	230
			– 5	200	220
			– 6	200	210
			– 7 or below	200	200